

Task-Conditioned Distributed Relational Formation Control Under a Global Formation Objective

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Abstract—This paper proposes a distributed control framework for multi-agent formation control under task-based and global objectives. To address the insensitivity of internal geometric structure to collective translation, we decouple the ideal formation into its internal geometry and the overall centroid position. This enables each agent to adjust its position using only local measurement and communication. The framework further incorporates each agent’s role-specific convex feasible region into the coupled system via projected local update. We establish sufficient conditions guaranteeing a unique equilibrium and geometric convergence of the coupled estimator-controller closed-loop system. Finally, numerical simulations on an 11-player soccer team—using distinct tactical diagrams and communication topologies—validate the framework: it successfully restores team formation without violating individual feasibility constraints, while ensuring convergence of both formation and centroid-estimation errors.

Index Terms—Formation control, multi-agent systems, distributed control, soccer tactics, distributed estimation

I. INTRODUCTION

In the research of formation control in multi-agent systems, what we usually focus on is not merely the specific positions of each agent, but rather the positional relationships between each agent and their “neighboring” agents. In a multi-agent system with internal collaboration mechanisms and communication, the formation cannot be simply represented by the absolute positions of each agent. The relative spatial relationships among different agents also contain important information about the internal structure and relative positions of the system. However, for the multi-agent system itself, its collective formation placement as a whole determines the position of the entire system. Therefore, when we require a multi-agent system to be capable of achieving formation control in complex situations and under the influence of various tasks, we must simultaneously consider both the internal geometric formation of the system and the collective formation placement. The tactical analysis of soccer team formations provides a natural example where the relational geometry of roles and the collective formation placement will change according to the task requirements [1], [2]. In this way, we can abstract the changes in the soccer team formation into a control problem.

The formation control of multi-agent systems is an important and widely studied topic, especially concerning the geometric structure within the system and the relative spatial relationships among different agents [3]–[5]. Graph-based formation-control methods provide a framework and tool

for encoding the geometric relations among agents [6], [7]. However, it is difficult to perceive and detect the collective translation of the entire system from relative relations. A common translation of all agents leaves all inter-agent relative displacements unchanged. Therefore, a multi-agent system may deviate from its expected collective formation placement, while still maintaining the required relational geometry. Therefore, when discussing the system’s “collective formation placement” simultaneously, considering only the analysis of “relational geometry” for the formation is not sufficient. This indicates that a complete multi-agent formation control requires considering both the internal relational geometry and the correctness of the collective formation placement.

Local relative measurements will provide information on the geometric relative positions of the neighbors, but they will not directly show the team centroid. Distributed estimation and regulation of collective quantities such as the centroid have been investigated in multi-agent systems. [8]–[10]. This limits the direct detection capability for the collective formation placement from local relative measurements. However, under the condition of the global formation objective of a specific task, the entire system needs to be in a correct collective formation placement, and the objective must regulate both quantities simultaneously. The current issue is how, in a local manner, we can utilize the distributed relational information as well as the distributed aggregation of the necessary collective information to accomplish the formation goals defined for the task at a global level.

We consider distributed formation execution under a fixed task condition and a fixed common formation reference. Both the task mode and the common formation reference are prescribed to all agents. We focus on the methods and feasibility in distributed execution. For this current framework, each specified task corresponds to a specific task-conditioned relational formation structure, which defines the relationships between agents within the system and the role-dependent feasible regions in geometric positions. Additionally, in the presence of a global formation objective, this requires the collective formation placement of the system to be correct and the accuracy of the relational geometry within the system and the relative positions of agents with different roles. For each agent, the team centroid is not directly available from local relative measurements; they will maintain a local centroid estimate and update it according to the information from communication with their neighbors [11].

This paper mainly contributes as follows:

- 1) We proposed a task-conditioned relational formation

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representation. This method can ensure the uniqueness of the posed formation when both collective translation and internal relational structure are considered, and also derives the orthogonal decomposition between the internal shape and collective translation.

- 2) We achieved the collective formation adjustment and control based on the team centroid by estimating the team centroid locally and through role-dependent convex projection.
- 3) For the coupled estimator-controller closed loop, we proved the coupled convergence guarantee of this coupled loop, and numerical simulations on a full 11-agent soccer formation illustrate the proposed framework.

II. PROBLEM FORMULATION AND RELATIONAL FORMATION MODEL

Considering that our multi-agent system is currently operating in a situation with a definite task, and there is also a relationship between the task and the global objective. Therefore, we will combine Task-Conditioned to re-encode the formation.

A. Task-Conditioned Formation Representation

Let $\mathcal{V} = \{1, \dots, n\}$ denote the agent set, where $n \geq 2$. The position of agent i at discrete time k is $p_i(k) \in \mathbb{R}^2$. At the tactical level, the agent dynamics are modeled as

$$p_i(k+1) = p_i(k) + hu_i(k), \quad (1)$$

where $h > 0$ is the sampling interval and $u_i(k) \in \mathbb{R}^2$ is the tactical motion command. We theoretically focused our attention on the design of $p_i(k+1)$, thereby the control input at time k is $u_i(k) = [p_i(k+1) - p_i(k)]/h$.

Define the stacked configuration $p = \text{col}(p_1, \dots, p_n) \in \mathbb{R}^{2n}$. Each agent is assigned a fixed tactical role ρ_i .

We consider a fixed task mode σ and a fixed common formation reference $\eta^r = (c^r, \psi^r, \ell^r)$ for every agent, where $c^r \in \mathbb{R}^2$ is the reference centroid, $\psi^r \in \mathbb{S}^1$ is the reference orientation, and $\ell^r = (\ell_{\parallel}^r, \ell_{\perp}^r) \in \mathbb{R}_{>0}^2$ specifies the longitudinal and lateral scales, it can be understood as the degree of compactness of the formation in both the longitudinal and lateral directions.

Now let's define the Reference Transformation:

$$T^r = R(\psi^r)S(\ell^r), \quad (2)$$

where $R(\psi^r)$ is the planar rotation matrix and $S(\ell^r) = \text{diag}(\ell_{\parallel}^r, \ell_{\perp}^r)$.

Throughout the paper, $\mathbf{1}_n$ denotes the n -dimensional all-ones vector, I_n denotes the $n \times n$ identity matrix, and $\otimes$ denotes the Kronecker product.

For the task mode σ , let $\bar{q}_i^{\sigma} \in \mathbb{R}^2$ denote the canonical slot of agent i , and define $\bar{q}^{\sigma} = \text{col}(\bar{q}_1^{\sigma}, \dots, \bar{q}_n^{\sigma})$, where $\text{col}(\cdot)$ denotes column stacking. The canonical template is centered such that:

$$(\mathbf{1}_n^{\top} \otimes I_2)\bar{q}^{\sigma} = 0. \quad (3)$$

The corresponding world-frame reference positions $q_i^r \in \mathbb{R}^2$ and stacked reference configuration $q^r \in \mathbb{R}^{2n}$ are

$$q_i^r = c^r + T^r \bar{q}_i^{\sigma}, \quad q^r = \mathbf{1}_n \otimes c^r + (I_n \otimes T^r)\bar{q}^{\sigma}. \quad (4)$$

Indeed, according to the meaning of the reference centroid c^r , $\frac{1}{n}(\mathbf{1}_n^{\top} \otimes I_2)q^r = c^r$.

For an arbitrary configuration p , define its actual team centroid as

$$c_p = \frac{1}{n}(\mathbf{1}_n^{\top} \otimes I_2)p.$$

Under this setting, $\bar{q}^{\sigma}$ determines the intrinsic geometric structure of the system during the execution of this task, while η^r determines the orientation and state of this intrinsic formation at the macroscopic level of the multi-agent system.

B. Relational Geometry and Collective Placement

Let $\mathcal{G}_f^{\sigma} = (\mathcal{V}, \mathcal{E}_f^{\sigma}, w^{\sigma})$ be the undirected task-conditioned formation graph [6], [7], where w^{σ} assigns a positive weight $w_{ij}^{\sigma} > 0$ to each edge $\{i, j\} \in \mathcal{E}_f^{\sigma}$. We assume that $\mathcal{G}_f^{\sigma}$ is connected.

We assign an arbitrary orientation to each edge of $\mathcal{G}_f^{\sigma}$ and let $B_{\sigma} \in \mathbb{R}^{n \times m_{\sigma}}$ be the corresponding incidence matrix, where $m_{\sigma} = |\mathcal{E}_f^{\sigma}|$, with -1 at the tail and $+1$ at the head of each oriented edge. For an oriented edge $i \rightarrow j$, define

$$\bar{r}_{ij}^{\sigma} = \bar{q}_j^{\sigma} - \bar{q}_i^{\sigma}, \quad r_{ij}^r = T^r \bar{r}_{ij}^{\sigma}. \quad (5)$$

At this point, we need to make a distinction and provide an explanation. $\bar{r}_{ij}^{\sigma}$ represents the desired displacement in the canonical frame, while r_{ij}^r represents the desired displacement after conversion in the world frame.

Let $r_{\sigma}^r \in \mathbb{R}^{2m_{\sigma}}$ denote the stacked world-frame desired edge displacement vector. Stacking all desired edge displacements gives:

$$r_{\sigma}^r = (B_{\sigma}^{\top} \otimes I_2)q^r. \quad (6)$$

Therefore, the desired relationships for edges are cycle-consistent automatically.

Proposition 1 (Relational shape uniqueness): If $\mathcal{G}_f^{\sigma}$ is connected, every configuration p satisfying

$$(B_{\sigma}^{\top} \otimes I_2)p = r_{\sigma}^r \quad (7)$$

can be written as

$$p = q^r + \mathbf{1}_n \otimes a, \quad a \in \mathbb{R}^2. \quad (8)$$

If, in addition, $c_p = c^r$, then $p = q^r$.

Proof: From (6) and (7),

$$(B_{\sigma}^{\top} \otimes I_2)(p - q^r) = 0.$$

Since $\mathcal{G}_f^{\sigma}$ is connected, $\ker(B_{\sigma}^{\top}) = \text{span}\{\mathbf{1}_n\}$. Because I_2 is nonsingular,

$$\ker(B_{\sigma}^{\top} \otimes I_2) = \{\mathbf{1}_n \otimes a : a \in \mathbb{R}^2\}.$$

Hence, $p - q^r = \mathbf{1}_n \otimes a$ for some $a \in \mathbb{R}^2$. Taking the centroid of both sides gives $c_p - c^r = a$. Therefore, if $c_p = c^r$, then $a = 0$ and $p = q^r$. This indicates that we can and only can find a unique formation shape, in which the relative positions of the agents satisfy the requirements of the geometric structure, and the actual formation centroid precisely coincides with the reference centroid. $\square$

To separate collective translation from internal shape, introduce the consensus and disagreement projectors:

$$P_{\parallel} = \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^{\top}, \quad P_{\perp} = I_n - P_{\parallel}. \quad (9)$$

Define the centered configurations:

$$p_{\perp} = (P_{\perp} \otimes I_2)p, \quad q_{\perp}^r = (I_n \otimes T^r)q^r.$$

By construction,

$$p = \mathbf{1}_n \otimes c_p + p_{\perp}, \quad q^r = \mathbf{1}_n \otimes c^r + q_{\perp}^r.$$

Thus,

$$p - q^r = \mathbf{1}_n \otimes (c_p - c^r) + (p_{\perp} - q_{\perp}^r).$$

The first term lies in the consensus subspace, whereas the second lies in the disagreement subspace by construction and (3); hence they are orthogonal. Since $\|\mathbf{1}_n \otimes v\|^2 = n\|v\|^2$, we obtain

$$\|p - q^r\|^2 = n\|c_p - c^r\|^2 + \|p_{\perp} - q_{\perp}^r\|^2. \quad (10)$$

From this, we can conclude that in this structure, the relation formation shape and the collective formation translation each occupy mutually orthogonal subspaces.

Moreover, since $B_{\sigma}^{\top} \mathbf{1}_n = 0$,

$$(B_{\sigma}^{\top} \otimes I_2)p = (B_{\sigma}^{\top} \otimes I_2)p_{\perp},$$

so the relational geometry is independent of collective translation.

For each agent, let $\bar{\mathcal{X}}_i^{\sigma} \subset \mathbb{R}^2$ be its canonical role-feasible set and let $\Omega \subset \mathbb{R}^2$ denote the admissible field domain. The corresponding world-frame feasible set is

$$\mathcal{X}_i^{\sigma}(\eta^r) = \Omega \cap (c^r + T^r \bar{\mathcal{X}}_i^{\sigma}). \quad (11)$$

Since η^r is fixed throughout the paper, we abbreviate $\mathcal{X}_i^{\sigma}(\eta^r)$ as $\mathcal{X}_i^{\sigma}$, and define

$$\mathcal{X}^{\sigma} = \prod_{i=1}^n \mathcal{X}_i^{\sigma}.$$

We assume Ω and $\bar{\mathcal{X}}_i^{\sigma}$ are nonempty, closed, and convex, also $\mathcal{X}_i^{\sigma} \neq \emptyset$, and $q^r \in \mathcal{X}^{\sigma}$. Since T^r is nonsingular, the affine image $c^r + T^r \bar{\mathcal{X}}_i^{\sigma}$ is closed and convex whenever $\bar{\mathcal{X}}_i^{\sigma}$ is closed and convex. Convex feasibility constraints are commonly incorporated in distributed multi-agent optimization and control through local constraint sets [12].

III. DISTRIBUTED RELATIONAL FORMATION CONTROL

A. Global Formation Objective

For each tactical relation, define the local relation formation error as: $e_{ij} = (p_j - p_i) - r_{ij}^r$. We should note that r_{ij}^r represents the desired displacement from i to j .

Let $W_{\sigma} \succ 0$ denote the diagonal matrix collecting the edge weights w_{ij}^{σ} according to the edge ordering associated with B_{σ} , and define the weighted formation Laplacian as

$$L_{\sigma} = B_{\sigma} W_{\sigma} B_{\sigma}^{\top}.$$

Based on the decomposition of p that we completed previously, at this stage, when constructing the objective function, the internal relational shape of the system and the collective translation can also be separated. We defined the shape cost J_s^{σ} and the centroid cost J_c as:

$$J_s^{\sigma}(p) = \frac{1}{2} \sum_{\{i,j\} \in \mathcal{E}_f^{\sigma}} w_{ij}^{\sigma} \|e_{ij}\|^2, \quad (12)$$

$$J_c(p) = \frac{n\kappa_c}{2} \|c_p - c^r\|^2, \quad \kappa_c > 0.$$

Using (6), the shape cost can equivalently be written as

$$J_s^{\sigma}(p) = \frac{1}{2} (p - q^r)^{\top} (L_{\sigma} \otimes I_2) (p - q^r).$$

At this point, we define the total objective function J^{σ} as the sum of the shape cost and the centroid cost:

$$J^{\sigma}(p) = J_s^{\sigma}(p) + J_c(p) = \frac{1}{2} (p - q^r)^{\top} H_{\sigma} (p - q^r),$$

$$H_{\sigma} = \left(L_{\sigma} + \frac{\kappa_c}{n} \mathbf{1}_n \mathbf{1}_n^{\top} \right) \otimes I_2 = (L_{\sigma} + \kappa_c P_{\parallel}) \otimes I_2. \quad (13)$$

Proposition 2 (J^{σ} Strong convexity and unique target):

Under the connectivity and positive-weight assumptions of $\mathcal{G}_f^{\sigma}$, $H_{\sigma} \succ 0$. Consequently, J^{σ} is strongly convex. Moreover, since $q^r \in \mathcal{X}^{\sigma}$, the constrained problem

$$\min_{p \in \mathcal{X}^{\sigma}} J^{\sigma}(p)$$

has the unique minimizer q^r .

Proof: For any $x \neq 0$, let $x_{\parallel} = (P_{\parallel} \otimes I_2)x$ and $x_{\perp} = (P_{\perp} \otimes I_2)x$. Since $L_{\sigma} \mathbf{1}_n = 0$ and the connectedness of $\mathcal{G}_f^{\sigma}$ implies that L_{σ} is positive definite on the disagreement subspace,

$$x^{\top} H_{\sigma} x = x_{\perp}^{\top} (L_{\sigma} \otimes I_2) x_{\perp} + \kappa_c \|x_{\parallel}\|^2 > 0.$$

Since $x = x_{\parallel} + x_{\perp} \neq 0$, the two components of x can't be both zero. Hence $H_{\sigma} \succ 0$. Since $q^r \in \mathcal{X}^{\sigma}$, $J^{\sigma}(q^r) = 0$, whereas $J^{\sigma}(p) > 0$ for every $p \neq q^r$. Therefore, q^r is the unique constrained minimizer. $\square$

The definitions of μ_{σ} and Λ_{σ} are as follows:

$$\mu_{\sigma} = \min\{\kappa_c, \lambda_2(L_{\sigma})\}, \quad \Lambda_{\sigma} = \max\{\kappa_c, \lambda_n(L_{\sigma})\}, \quad (14)$$

so that $\mu_{\sigma} I_{2n} \preceq H_{\sigma} \preceq \Lambda_{\sigma} I_{2n}$.

We denote $\mathcal{N}_i^f$ as the neighbor of agent i in the tactical formation graph. $\mathcal{N}_i^f := \{j : \{i, j\} \in \mathcal{E}_f^{\sigma}\}$.

Take the local gradient of the total cost J^{σ} , we have $\nabla_{p_i} J_s^{\sigma} = -\sum_{j \in \mathcal{N}_i^f} w_{ij}^{\sigma} e_{ij}$, $\nabla_{p_i} J_c = \kappa_c (c_p - c^r)$. Hence, the ideal local gradient is

$$g_i^{\text{id}} := \nabla_{p_i} J^{\sigma}(p) = -\sum_{j \in \mathcal{N}_i^f} w_{ij}^{\sigma} e_{ij} + \kappa_c (c_p - c^r). \quad (15)$$

The relational term in (15) is locally computable from relative measurements and stored task-conditioned relation data. On the contrast, the actual centroid c_p is a collective quantity that cannot be obtained from the prescribed local measurements.

B. Distributed Realization

At this point, under the "fixed task" scenario, we make the following assumption: all agents share a common world coordinate frame and update synchronously. For compactness, in this paper x^+ denotes the next-step value of x .

Let $\mathcal{G}_s(k)$ denote the sensing graph, and let $\mathcal{E}_s^b(k)$ denote the set of agent pairs that apply bidirectional noiseless mutual sensing at time k . We assume that every active tactical edge is supported by such sensing, i.e.,

$$\mathcal{E}_f^\sigma \subseteq \mathcal{E}_s^b(k), \quad \forall k.$$

For each $(i, j) \in \mathcal{E}_s^b(k)$, the relative measurement is

$$\delta_{ij}(k) = p_j(k) - p_i(k).$$

Hence, for every tactical neighbor $j \in \mathcal{N}_i^f$, the local relation error $e_{ij} = \delta_{ij} - r_{ij}^r$ is directly available to agent i .

Let $\mathcal{G}_c = (\mathcal{V}, \mathcal{E}_c, A_c)$ be a fixed, undirected, and connected communication graph, where $A_c = [a_{ij}]$ and L_c denote its adjacency and Laplacian matrices, respectively. Define the communication-neighbor set as $\mathcal{N}_i^c = \{j : \{i, j\} \in \mathcal{E}_c\}$.

For a nonempty closed convex set C , let $\Pi_C(x) = \arg \min_{y \in C} \frac{1}{2} \|y - x\|^2$ denote the Euclidean projection, which is nonexpansive [12].

The actual collective centroid c_p of the system is not directly available from local relative measurements.. Therefore, for agent i , it will use $\hat{c}_i$ as the estimate of the actual centroid. We define the actual distributed gradient as:

$$\begin{aligned} g_i^{\text{dist}} &= - \sum_{j \in \mathcal{N}_i^f} w_{ij}^\sigma e_{ij} + \kappa_c (\hat{c}_i - c^r), \\ p_i^+ &= \Pi_{\mathcal{X}_i^\sigma} [p_i - \alpha g_i^{\text{dist}}], \quad \alpha > 0. \end{aligned} \quad (16)$$

We employ a dynamic-average consensus centroid estimator [11]. When agent i completes the transition $p_i \rightarrow p_i^+$, it will update the estimator:

$$\hat{c}_i^+ = \hat{c}_i - \gamma \sum_{j \in \mathcal{N}_i^c} a_{ij} (\hat{c}_i - \hat{c}_j) + (p_i^+ - p_i), \quad (17)$$

Here, $\gamma > 0$ represents the estimator communication gain.

We denote the stacked estimation of centroid: $\hat{c} = \text{col}(\hat{c}_1, \dots, \hat{c}_n)$. The initialization is required to satisfy:

$$\frac{1}{n} (\mathbf{1}_n^\top \otimes I_2) \hat{c}(0) = c_p(0). \quad (18)$$

A simple local initialization satisfying (18) is $\hat{c}_i(0) = p_i(0)$ for all i .

Lemma 1 (Estimator average preservation): Under (18),

$$\frac{1}{n} \sum_{i=1}^n \hat{c}_i(k) = c_p(k), \quad \forall k.$$

Proof: Summing (17) over all agents and using $\mathbf{1}_n^\top L_c = 0$ gives

$$\frac{1}{n} \sum_{i=1}^n \hat{c}_i^+ = \frac{1}{n} \sum_{i=1}^n \hat{c}_i + (c_p^+ - c_p).$$

The result follows from (18) by induction. $\square$

We define the estimation error $\tilde{c}_i$ as $\tilde{c}_i = \hat{c}_i - c_p$, and its stacked form $\tilde{c} = \text{col}(\tilde{c}_1, \dots, \tilde{c}_n)$. We also define the zero average estimator error subspace $\mathcal{D}_c = \{z \in \mathbb{R}^{2n} : (\mathbf{1}_n^\top \otimes I_2) z = 0\}$. From Lemma 1, we know $\tilde{c}(k) \in \mathcal{D}_c$. Define $C_\gamma = I_n - \gamma L_c$. For the communication gain γ and the communication disagreement contraction β_c :

$$0 < \gamma < \frac{2}{\lambda_n(L_c)}, \quad \beta_c = \max_{i=2, \dots, n} |1 - \gamma \lambda_i(L_c)| < 1. \quad (19)$$

Consequently, since communication graph is connected, L_c is symmetric, for every $z \in \mathcal{D}_c$,

$$\|(C_\gamma \otimes I_2) z\| \leq \beta_c \|z\|.$$

Define $g^{\text{dist}} = \text{col}(g_1^{\text{dist}}, \dots, g_n^{\text{dist}})$.

In a connected undirected graph, it can be guaranteed that $0 = \lambda_1(L_c) < \lambda_2(L_c)$. Since $\hat{c}_i = c_p + \tilde{c}_i$, we have:

$$g_i^{\text{dist}} = g_i^{\text{id}} + \kappa_c \tilde{c}_i, \quad g^{\text{dist}} = \nabla J^\sigma(p) + \kappa_c \tilde{c}. \quad (20)$$

Since $\mathcal{X}^\sigma = \prod_{i=1}^n \mathcal{X}_i^\sigma$, the Euclidean projection onto $\mathcal{X}^\sigma$ decomposes componentwise across the agents.

Hence, we can have the stacked position update as:

$$p^+ = \Pi_{\mathcal{X}^\sigma} [p - \alpha (H_\sigma(p - q^r) + \kappa_c \tilde{c})]. \quad (21)$$

According to the estimator and position update equations we proposed, at this point, the controller only requires the current position p_i of agent i , its relative measurements δ_{ij} of the neighbors, its own stored task-conditioned local data, common (σ, η^r) , and the estimated value $\hat{c}_i$ of the real-time centroid c_p by its estimator. The controller does not need to directly access the actual c_p , the complete system collective configuration p , or the global matrices L_σ, H_σ . And merely relying on distributed estimation, as well as its own knowledge base $\mathcal{K}_i^\sigma$, which is $\mathcal{K}_i^\sigma = \{\rho_i, \mathcal{N}_i^f, (w_{ij}^\sigma, \bar{r}_{ij}^\sigma)_{j \in \mathcal{N}_i^f}; \bar{\mathcal{X}}_i^\sigma\}$. The synchronous implementation follows *sense* $\rightarrow$ *control/project/move* $\rightarrow$ *estimator update*; hence the estimator in (17) uses the realized position increment $p_i^+ - p_i$ and current neighbor estimates, without introducing an algebraic loop.

IV. CONVERGENCE ANALYSIS

In this section, we establish the geometric convergence of the coupling "formation and estimator" closed-loop system.

Define the formation error $e_p = p - q^r$ and the scalar error measures $\epsilon_p = \|e_p\|$ and $\epsilon_c = \|\tilde{c}\|$.

Using $C_\gamma \mathbf{1}_n = \mathbf{1}_n$ and

$$\mathbf{1}_n \otimes (c_p^+ - c_p) = (P_\parallel \otimes I_2)(p^+ - p),$$

The stacked estimation error satisfies

$$\tilde{c}^+ = (C_\gamma \otimes I_2) \tilde{c} + (P_\perp \otimes I_2)(p^+ - p). \quad (22)$$

For the formation subsystem, define

$$\beta_\sigma(\alpha) = \max \{|1 - \alpha \mu_\sigma|, |1 - \alpha \Lambda_\sigma|\}.$$

Since H_σ is symmetric positive definite and $\mu_\sigma I_{2n} \preceq H_\sigma \preceq \Lambda_\sigma I_{2n}$,

$$\|I_{2n} - \alpha H_\sigma\|_2 = \beta_\sigma(\alpha).$$

Theorem 1 (Formation convergence): Under the standing assumptions in Sections II and III, suppose $p(0) \in \mathcal{X}^\sigma$ and the estimator is initialized according to (18). If

$$0 < \alpha < \min \left\{ \frac{1}{\Lambda_\sigma}, \frac{\mu_\sigma(1 - \beta_c)}{\kappa_c(\Lambda_\sigma + \mu_\sigma)} \right\}, \quad (23)$$

then $\mathcal{Z} = \mathcal{X}^\sigma \times \mathcal{D}_c$ is forward invariant and, in the $(p, \tilde{c})$ coordinates, $(q^r, 0)$ is its unique equilibrium, and there exist $K > 0$ and $\varrho \in (0, 1)$ such that

$$\|p(k) - q^r\| + \|\tilde{c}(k)\| \leq K\varrho^k (\|p(0) - q^r\| + \|\tilde{c}(0)\|).$$

Consequently, $p(k) \rightarrow q^r$ and $\hat{c}_i(k) - c_p(k) \rightarrow 0$ for all i . This indicates that ultimately, the estimated value of the system centroid by agent i , $\hat{c}_i$, will be consistent with the actual centroid of the system c_p , and at this time, the relational geometric configuration within the system p will also be consistent with the reference geometric configuration q^r .

Proof: Since $p(0) \in \mathcal{X}^\sigma$ and the controller applies the projection onto $\mathcal{X}^\sigma$, we have $p(k) \in \mathcal{X}^\sigma$ for all k . Lemma 1 also gives $\tilde{c}(k) \in \mathcal{D}_c$; hence $\mathcal{Z}$ is forward invariant.

Moreover, at $p = q^r$ and $\tilde{c} = 0$, the distributed gradient vanishes; hence (21) gives $p^+ = q^r$. Then (22) gives $\tilde{c}^+ = 0$. Thus $(q^r, 0)$ is an equilibrium. We first conduct an error analysis of the formation subsystem. By projection nonexpansiveness and $\Pi_{\mathcal{X}^\sigma}(q^r) = q^r$,

$$\begin{aligned} \epsilon_p^+ &\leq \|(I_{2n} - \alpha H_\sigma)e_p - \alpha \kappa_c \tilde{c}\| \\ &\leq \beta_\sigma(\alpha)\epsilon_p + \alpha \kappa_c \epsilon_c. \end{aligned} \quad (24)$$

Hence, we can show that the formation subsystem error is upper bounded.

For the estimator subsystem, since $\tilde{c}(k) \in \mathcal{D}_c$ by Lemma 1, from (22) and the disagreement contraction in (19),

$$\epsilon_c^+ \leq \beta_c \epsilon_c + \|p^+ - p\|.$$

Since $p \in \mathcal{X}^\sigma$, $\Pi_{\mathcal{X}^\sigma}(p) = p$, and projection nonexpansiveness gives

$$\|p^+ - p\| \leq \alpha \Lambda_\sigma \epsilon_p + \alpha \kappa_c \epsilon_c.$$

Therefore,

$$\epsilon_c^+ \leq \alpha \Lambda_\sigma \epsilon_p + (\beta_c + \alpha \kappa_c) \epsilon_c. \quad (25)$$

From the above discussion of the classification of errors, it can be seen that both formation error ϵ_p and estimator error ϵ_c are upper-bounded. We reformulated it in a more compact form, and compressed the errors dynamics into a two-dimensional system for analysis:

Combining (24) and (25) yields the componentwise comparison system

$$z^+ \leq A_\alpha z, \quad A_\alpha = \begin{bmatrix} \beta_\sigma(\alpha) & \alpha \kappa_c \\ \alpha \Lambda_\sigma & \beta_c + \alpha \kappa_c \end{bmatrix}, \quad z = \begin{bmatrix} \epsilon_p \\ \epsilon_c \end{bmatrix}. \quad (26)$$

Under (23), $\alpha < 1/\Lambda_\sigma$, and hence $\beta_\sigma(\alpha) = 1 - \alpha \mu_\sigma$. Moreover, since $\mu_\sigma/(\Lambda_\sigma + \mu_\sigma) < 1$, (23) implies $\alpha \kappa_c < 1 - \beta_c$, and hence $[A_\alpha]_{22} = \beta_c + \alpha \kappa_c < 1$. For the nonnegative

2×2 matrix A_α , the standard Schur conditions are satisfied since $[A_\alpha]_{11} < 1$, $[A_\alpha]_{22} < 1$, and

$$\alpha \kappa_c (\Lambda_\sigma + \mu_\sigma) < \mu_\sigma (1 - \beta_c),$$

where the last inequality is same as $[A_\alpha]_{12}[A_\alpha]_{21} < (1 - [A_\alpha]_{11})(1 - [A_\alpha]_{22})$. Thus $\rho(A_\alpha) < 1$.

Since $A_\alpha \geq 0$, iteration of (26) gives $z(k) \leq A_\alpha^k z(0)$ componentwise. Because $\rho(A_\alpha) < 1$, A_α^k decays geometrically, which clearly proves the stated convergence bound. Any other equilibrium in $\mathcal{Z}$ would generate a constant trajectory and hence contradict convergence to $(q^r, 0)$. Therefore, $(q^r, 0)$ is the unique equilibrium. $\square$

V. SIMULATION

Multi-agent formation control has broad and essential application in soccer strategy and modeling analysis [1], [2]. We consider a 11-player soccer team, and the formation used during the fixed task is 4-2-3-1. We should notice the task mode σ is externally prescribed and remains fixed throughout the simulation. And we define the field is $\Omega = [0, 105] \times [0, 68]$ m², with the attacking direction along the positive x -axis. During this fixed task, the common formation reference is $c^r = [45, 34]^T$ m, $\psi^r = 0$, and $\ell^r = [10, 8]^T$. A centered canonical 4-2-3-1 template satisfying $(\mathbf{1}_{11}^T \otimes I_2)\bar{q}^\sigma = 0$ is used to generate the posed reference configuration q^r .

The initial formation configuration is distorted from the desired formation while remaining in the feasible region, with $p(0) \in \mathcal{X}^\sigma$ and $\|p(0) - q^r\|_2 = 32.0$ m.

Each player is assigned a rectangular role-feasible region $\mathcal{X}_i^\sigma$, as shown in Fig. 1(a). Under this background setting, starting from a significant initial deviation, the team's formation recovery and the two information graphs (tactical formation and communication) used in the simulation are shown in Fig. 1.

The fixed formation graph $\mathcal{G}_f^\sigma$ contains 19 positively weighted edges and is connected, with $\lambda_2(L_\sigma) = 0.91519$ and $\lambda_{11}(L_\sigma) = 8.48462$. Every valid tactical edge can apply accurate mutual relative sensing, so that $\mathcal{E}_f^\sigma \subseteq \mathcal{E}_s^b(k)$ for all k . The communication graph $\mathcal{G}_c$ contains 25 unit-weight edges, is fixed, undirected, and connected, and satisfies $\mathcal{G}_c \neq \mathcal{G}_f^\sigma$. Its extreme nonzero Laplacian eigenvalues in this simulation setting are $\lambda_2(L_c) = 1.64481$ and $\lambda_{11}(L_c) = 8.35582$.

The centroid estimator is initialized locally with $\hat{c}_i(0) = p_i(0)$, which satisfies the average-consistency condition in (18). The gains of the controller and estimator that we have chosen are: $\kappa_c = 1$, $\gamma = 0.20$, and $\alpha = 0.020$. From (14), in this scenario, $\mu_\sigma = 0.91519$ and $\Lambda_\sigma = 8.48462$. For $\gamma = 0.20$, the communication graph gives $\beta_c = 0.67116$. Moreover, $\gamma = 0.20 < 2/\lambda_{11}(L_c) \approx 0.23935$, so the estimator contraction condition in (19) is satisfied. Defining $\alpha_{\max}^{\text{suff}}$ as the right-hand side of (23), the present parameters yield $\alpha_{\max}^{\text{suff}} = 0.032016$. Hence, $\alpha = 0.020 < \alpha_{\max}^{\text{suff}}$.

As shown in Fig. 1(a), all agents recover their prescribed reference positions while keep remaining within their feasible regions, they also managed to keep the formation centroid and the reference centroid in alignment very well.

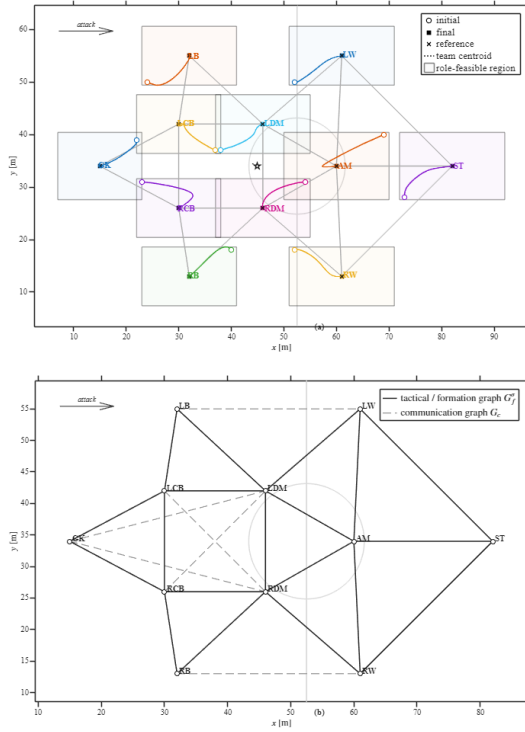


Fig. 1. Nominal 11-agent task-conditioned relational formation simulation. (a) Full-team formation recovery from a deliberately distorted but feasible initial configuration; shaded rectangles denote the role-feasible regions. (b) Superposed tactical/formation and communication graphs, where solid and dashed edges denote G_f^σ and G_c , respectively.

Fig. 1(b) illustrates the distinct tactical and communication architectures, where the solid and dashed edges represent G_f^σ and G_c , respectively.

The convergence behavior of the two scalar errors appearing in the comparison system (26) is shown in Fig. 2. Based on the decay of the formation error and the centroid estimation error in the numerical simulation, this is consistent with the convergence guarantee in our Theorem 1.

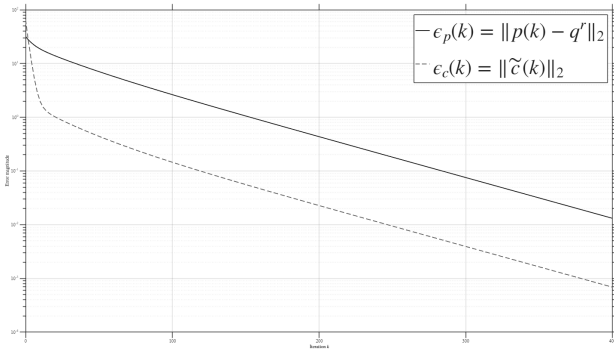


Fig. 2. Closed-loop convergence of the formation error $e_p(k) = \|p(k) - q^*\|_2$ and centroid-estimation error $e_c(k) = \|\tilde{c}(k)\|_2$ over the first 400 iterations. Both errors are shown on a logarithmic scale.

VI. CONCLUSIONS

This article presents a formation control framework for distributed multi-agent systems in the case where there is a task-conditioned and global formation objective. We re-described this formation control problem by separating the desired configuration into its internal relational geometry and collective placement. Through this task-conditioned relational representation, we have constructed a distributed action change strategy that enables each agent to update its estimation of the actual centroid based on the neighbor communication and its own realized motion. The role-dependent convex feasible sets are also integrated with this to ensure the feasibility of position changes. Under the stated connectivity, Considering sensing, feasibility and gain conditions, the coupled estimator-controller closed-loop system under this framework has a unique equilibrium point (reference formation) and converges geometrically. We later demonstrated our theory through the formation-estimator error convergence and full team formation recovery of the 11-player soccer team. In the future, we will extend the framework to task-switching scenarios and event-triggered multi agent formation control.

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